

Project Execution Tasks & Hackathon Milestones

Project: Project Rudra-Logistics (SIH Problem ID 26251)

Status: In-Progress / Initializing

Tracking System: Checklist format with dependencies and ownership tags

Phase 1: Foundation & Data Generation (Sprint 1)

- [x] **Task 1.1: Project Scaffolding**
 - Initialized `Backend` with FastAPI, Uvicorn, and requirements.txt.
 - Initialized `Frontend` with React (Vite) and TailwindCSS.
 - Configured CORS, environment variables, and tactical dark theme config.
 - *Status: Completed*
 - [x] **Task 1.2: Military Sector Geo-Spatial Dataset**
 - Defined exact coordinates and altitudes for Northern Command corridor (Leh, South Pullu, Khardung La, North Pullu, Partapur, Siachen Base Camp, DBO).
 - Implemented primary route and alternate bypass route (via Shyok river axis).
 - *Status: Completed*
 - [x] **Task 1.3: Synthetic Military Consumption Data Generator**
 - Created Python script `data_generator.py` producing 180 days of historical records for Class I (Rations), Class III (POL Kerosene & Diesel), Class V (Ammunition), and Class VIII (Medical).
 - Injected realistic seasonality with sub-zero winter temperatures.
 - *Status: Completed*
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Phase 2: AI Forecasting Engine & Inventory Logic (Sprint 2)

- [x] **Task 2.1: Multi-Factor Demand Forecasting Model**

- Built Python module `forecast_service.py` .
 - Implemented time-series prediction incorporating base consumption, sub-zero temperature non-linear heating decay, and threat level multipliers.
 - *Status: Completed*
 - [x] **Task 2.2: Days of Supply (DOS) Calculation Engine**
 - Implemented automatic DOS formula: $\text{DOS} = \frac{\text{Current Inventory}}{\text{Forecasted Daily Consumption}}$.
 - Categorized status into RED (≤ 5 days), AMBER (5-15 days), GREEN (> 15 days).
 - Exposed API: `GET /api/v1/nodes/health` .
 - *Status: Completed*
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Phase 3: Dynamic GIS Routing & Roadblock Engine (Sprint 3)

- [x] **Task 3.1: Network Graph & Routing Solver**
 - Built graph representation using Python `NetworkX` .
 - Calculated travel times, fuel penalties, and altitude gradient strain.
 - *Status: Completed*
 - [x] **Task 3.2: Dynamic Hazard & Pass Closure Simulator**
 - Implemented API endpoint: `POST /api/v1/passess/toggle` .
 - Dynamic discovery of alternate bypass route via Agham-Shyok axis when Khardung La is blocked (+4.2 hrs delay, +160 L fuel).
 - Exposed API: `POST /api/v1/routes/optimize` .
 - *Status: Completed*
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Phase 4: IoT Telemetry & Telemetry Streaming (Sprint 4)

- [x] **Task 4.1: Live Convoy Telemetry Simulator**
- Built `telemetry_service.py` simulating moving convoy Stallion-Alpha.
- Broadcast payload over FastAPI WebSocket `/ws/telemetry` with fallback REST endpoint.
- *Status: Completed*
- [x] **Task 4.2: Automated Indenting & Cold-Chain Alert Triggers**

- Generated automatic alert payload if cargo temperature drifts outside 2°C - 8°C.
 - Generated automated replenishment indent when any post hits $\text{DOS} \leq 7$ days.
 - Exposed API: `POST /api/v1/nodes/indents/authorize`.
 - *Status: Completed*
 - [x] **Task 4.3: Cryptographic Audit Chain & Anti-Spoofing Engine**
 - Implemented SHA-256 hash chaining module for immutable transaction logging with Genesis block.
 - Built kinematic anomaly detector in telemetry pipeline flagging velocity jumps > 90 km/h or teleportation.
 - *Status: Completed*
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Phase 5: Tactical C2 Frontend Dashboard (Sprint 5)

- [x] **Task 5.1: Master Layout & Header HUD**
- Military command dark container with live Zulu/IST clock, threat status toggle, and alert marquee.
- *Status: Completed*
- [x] **Task 5.2: Interactive Tactical GIS Map**
- Custom dark tactical SVG map with topography, outposts, depots, pass blockage icons, and moving convoy icon.
- Visual route switching when roadblock is toggled.
- *Status: Completed*
- [x] **Task 5.3: DOS Stock Health Grid & Cards**
- Responsive card matrix for Siachen Base Camp, DBO, Turtuk, and Partapur.
- Color-coded progress bars for Kerosene, Ammo, Rations, and Medical supplies.
- Quick action `AUTHORIZE DISPATCH` modal with cryptographic signing.
- *Status: Completed*
- [x] **Task 5.4: "What-If" War-Gaming Sandbox Component**
- Interactive slider controls for temperature (-5°C to -45°C), troop mobilization (+0% to +100%), and pass blockade (0 to 14 days).
- Dynamic forward depletion charts and AI recommendations.
- *Status: Completed*

- [x] **Task 5.5: Master Admin Operational Console UI**
 - Global multiplier adjustments, emergency stock injection form, and operational controls.
 - *Status: Completed*
 - [x] **Task 5.6: Cyber Defense HUD & Tamper Explorer**
 - Tamper verification shield with audit chain explorer.
 - One-click `SIMULATE TAMPER ATTACK` and `RE-SIGN CHAIN` demonstration features.
 - *Status: Completed*
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Phase 6: Polish, Air-Gapped Packaging & Hackathon Pitch (Sprint 6)

- [x] **Task 6.1: Air-Gapped Packaging**
- Completely local offline bundle without external CDN or Google Fonts leaks.
- Synthesized Web Audio radar pings with zero external audio assets.
- *Status: Completed*